# Kishore Kaarthik

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## Professional Summary

Entry-level software and security engineer building secure Python backends and ML systems. Built a multi-tenant PII vault (FastAPI, PostgreSQL row-level security, AES-256-GCM, 410+ automated tests) and a multimodal DistilBERT + ResNet-18 abuse detector with a Chrome extension, presented at ICACTEA-2026. Fortinet certified (FortiGate, FortiManager), working in Python, Java and TypeScript.

## Education

**Rajalakshmi Engineering College, Anna University** — Chennai, India

*B.E. Computer Science and Engineering, CGPA 7.75* — *Nov 2022 – May 2026*

## Projects

**SecureDB Vault** | *Python, FastAPI, PostgreSQL, SQLAlchemy, AES-256-GCM, Argon2id*

- **Situation/Task:** The DPDP Act 2023 requires strict protection of PAN, Aadhaar and bank data
- **Action:** Built a multi-tenant PII tokenization REST API with per-tenant AES-256-GCM envelope encryption (master key in the OS keychain, ciphertext bound to tenant, token and key version), and enforced tenant isolation with forced PostgreSQL row-level security and least-privilege column-level grants
- **Action:** Added a hash-chained tamper-evident audit log with a verification endpoint, deny-by-default per-API-key policies (allow, mask, deny) with rate limiting, DPDP erasure and export, and zero-downtime key rotation
- **Result:** Verified with 410+ automated tests (unit, property-based, concurrency, security-regression) run against a real PostgreSQL instance

**CyberShield** | *PyTorch, DistilBERT, ResNet-18, Flask, Chrome Extension (MV3), X API*

- **Situation/Task:** Text-only moderation misses image and emoji subtext and flags abuse after posting
- **Action:** Built a multimodal PyTorch model fusing DistilBERT, ResNet-18 and an emoji embedding through a transformer encoder into a 6-class abuse classifier plus aggression score, trained on MMHS150K with majority-vote labels
- **Action:** Built an explainable Flask detection engine with weighted lexicon rules, anti-evasion text normalization, risk zones, and a Chrome extension for X with a live toxicity meter and rewrite suggestions, plus account-level scoring via the X API
- **Result:** Co-authored with Kiruthikrokan B and presented the paper "A Cross-Modal Transformer for Cyberbullying Detection on X" at ICACTEA-2026, the 3rd International Conference on Algorithms and Computational Theory for Engineering Applications (Aditya University, March 2026)

## Certifications and Training

**Fortinet:** Certified Professional (Secure Networking), NSE 1–6, FortiGate 7.4 Administrator & Operator, FortiManager 7.4 Administrator

**Google Cloud:** GenAI Apps with Gemini and Streamlit, Vertex AI Prompt Design, API Gateway

**AWS Academy** Cloud Architecting

**CareerTiq Cyber Security Training** (114 hours, 2025): network security, VAPT and OWASP Top 10, SIEM and incident response, IAM and Active Directory, with TryHackMe labs

## Awards

**Winner** – Smart India Hackathon, Internal Round (2025); **Competitor** – Hackinfinity Inter-College Hackathon

## Technical Skills

**Languages**: Python, Java, JavaScript/TypeScript, SQL, Bash
**Security**: AES-256-GCM, Argon2id, row-level security, OAuth2/JWT, RBAC, OWASP Top 10, Fortinet FortiGate/FortiManager, firewalls, VPN, TCP/IP, SIEM and incident response fundamentals, Active Directory, Kali Linux, Nmap, Burp Suite
**Development**: FastAPI, Flask, REST APIs, PostgreSQL, MongoDB, MySQL, Oracle, React, Git, Docker, Linux, AWS, Google Cloud
**AI/ML**: PyTorch, DistilBERT, ResNet-18, Gemini API, prompt engineering, Claude Code